



Waste Heat Recovery based Desalination for Power Plants: The waste heat from power plant discharge can be tapped to produce desalinated water using the LTTD process. NIOT has successfully commissioned a LTTD plant in the North Chennai Thermal Power Station, Chennai. The major components from the 1000m³/day barge mounted desalination plant have been used to install the LTTD plant at NCTPS.

Wave energy conversion: The group is also committed to work on ocean based renewable energy schemes. As a part of it, the group has also commissioned the first ever wave powered desalination plant at Vizhinjam, India. This wave energy plant is one of the very few working wave energy plants in the world. The energy produced from this plant is being used to run a reverse osmosis based 10m³/day desalination plant to cater to the needs of the local fishing community. The current working version that has been operational for the past ten years has an impulse turbine driven by an oscillating water column. The earlier experiments with the plant configuration included a constant chord Wells turbine model and also a variable chord Wells turbine. The group is currently focusing on floating wave powered devices with potential applications in remote islands. One such device, a Backward Bent Ducted Buoy is currently being installed and tested at NIOT.

Ocean Thermal Energy Conversion: The OTEC uses the ocean thermal gradient to produce electricity by using a suitable refrigerant in a Rankine cycle. Riding on the success of its desalination program, the group has active plans to seriously pursue the Ocean Thermal Energy Conversion (OTEC) program in order to make the desalination plant self-sufficient. NIOT is in the process of installing an OTEC module in one of its island desalination plants.

- Develop turbines for OTEC/Wave Energy to run LTTD plants or run small loads or increase efficiency of wave energy plants.
- Conduct theoretical and experimental studies on components like heat exchangers for desalination and energy conversion.
- Install a floating Wave powered device to power loads like a small rated Reverse Osmosis or to meet the lighting requirements of small islands
- Generate theoretical, experimental and field data, for optimization of components and suitable adoption towards commercialization of extraction of renewable energies and fresh water including solar desalination.

SUCCESSFUL VENTURES:

- 50,000 LPD Floating desalination plant was successfully demonstrated for few weeks at a mid-sea off Tuticorin port in the year 2005.
- One lakh LPD land based LTTD plant was successfully installed and commissioned at Kavaratti in the year 2005. It has been supplying fresh water to the local community for the past three years the past four years.
- One MLD Barge Mounted Desalination Plant was successfully demonstrated for few weeks at mid-sea (sea depth of 1000m), off Ennore port in the year 2007.
- The wave powered desalination plant was successfully installed and commissioned at Vizhinjam, Kerala.

Current Endeavors: NIOT is in the process of installation of LTTD plants in three more islands in Lakshadweep one each at Agatti, Androth and Minicoy. The commissioning is expected to be around December 2009. The design of a larger capacity 10 MLD offshore plant is also in progress. Efforts are on to find suitable industry partners for transfer of technology.